Sparse AD

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☐ Full Width Mode ☐ Present Mode

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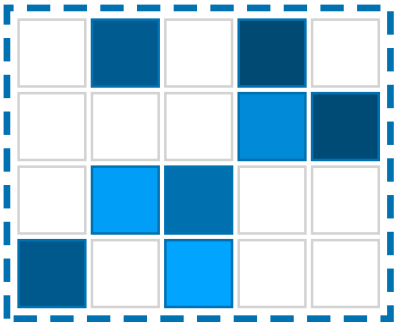
- What color is your Jacobian ? Graph Coloring for Computing Derivatives
- An Illustrated Guide to Automatic Sparse Differentiation
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Sparse Jacobian

Suppose the Jacobian is **sparse**.

- The *sparsity pattern* (rows and columns of nonzeros) is **known**
- You want to determine the **value** of these nonzeros with AD

Consider the 4×5 sparse matrix on the right:

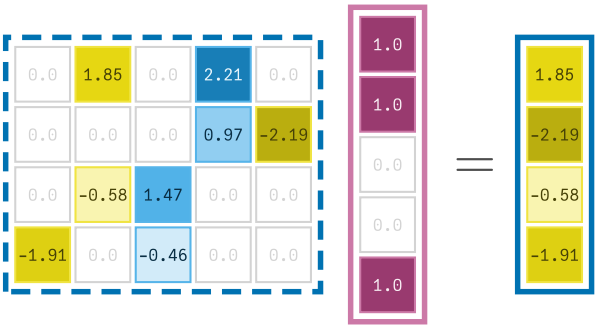


- Forward mode would require 5 JVP as there are 5 columns
- Reverse mode would require 4 VJP as there are 4 rows

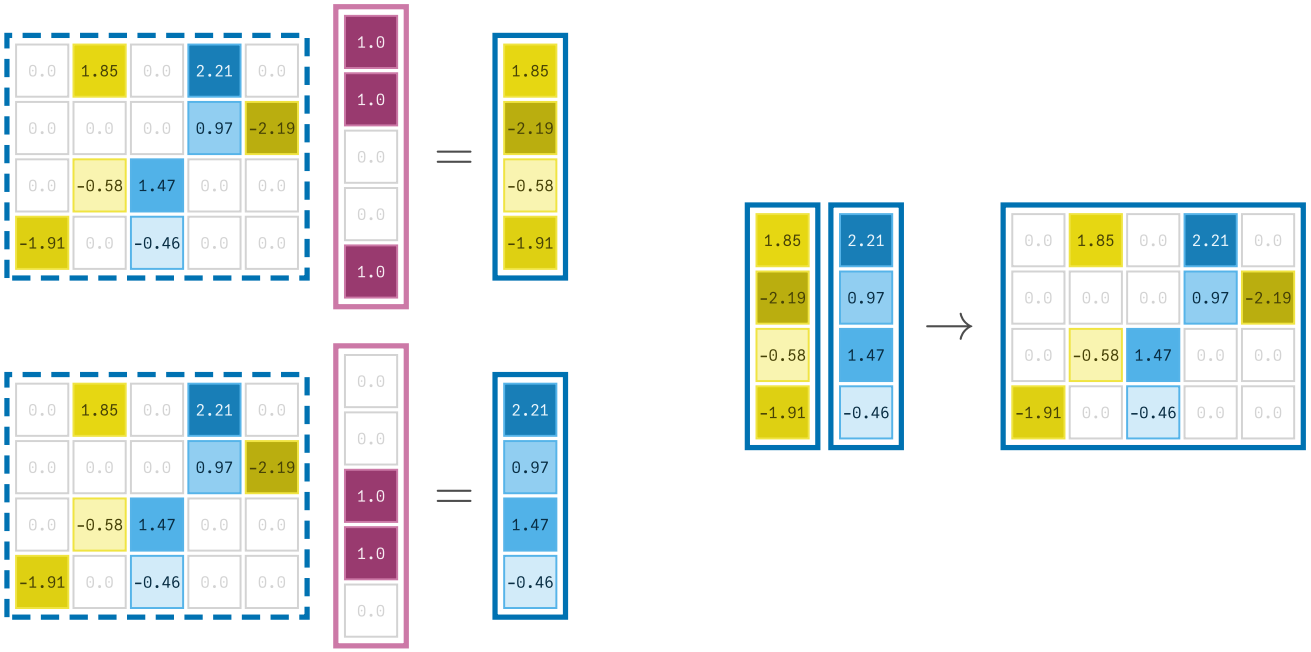
Can we do better using the known sparsity pattern ?

Three columns in just one JVP

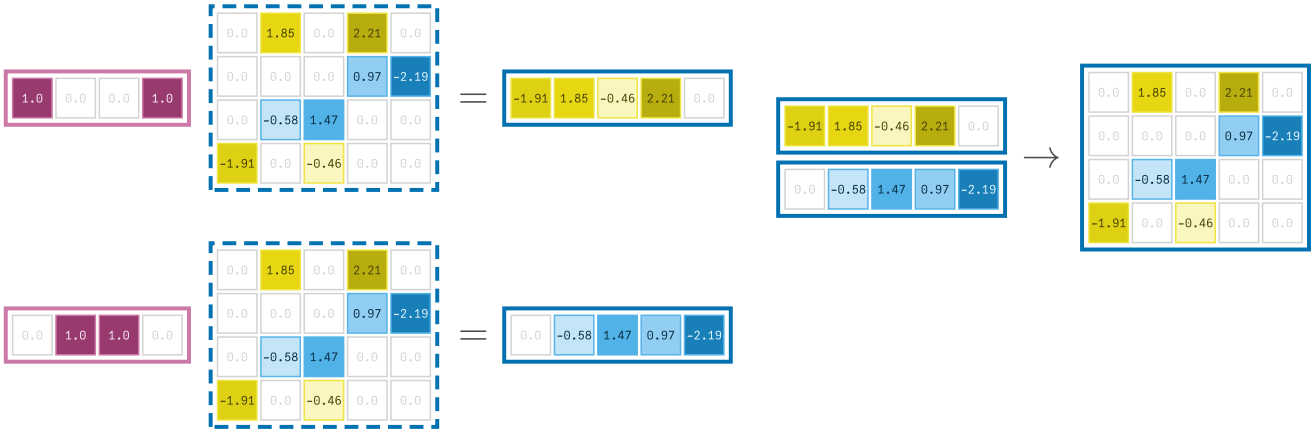
- The columns 1, 2 and 5 do not share any rows with nonzero entries.
- So their entries can be recovered unambiguously with just one JVP!



Whole Jacobian in just two JVP



The same applies to reverse mode



Sparsity detection of Jacobian

Simple operator overloading implementation

```
struct MyGradientTracer
    indexset::Set{Int}
end
```

```
Base.:+(a::MyGradientTracer, b::MyGradientTracer) =
MyGradientTracer(union(a.indexset, b.indexset))
```

```
Base.:*(a::MyGradientTracer, b::MyGradientTracer) =
MyGradientTracer(union(a.indexset, b.indexset))
```

```
Base.:/(a::MyGradientTracer, b::Number) = a
```

The sign function is constant over $x > 0$ and $x < 0$ so its output carries **no** variables

```
Base.sign(x::MyGradientTracer) = MyGradientTracer(Set()) # return empty index set
```

Scalar example

```
f (generic function with 1 method)
```

```
f(x) = x[1] + x[2]*x[3] * sign(x[4])
```

```
xtracer =
▸ [MyGradientTracer(Set([1])), MyGradientTracer(Set([2])), MyGradientTracer(Set([3])), MyGradie
```

```
xtracer = [
    MyGradientTracer(Set(1)),
    MyGradientTracer(Set(2)),
    MyGradientTracer(Set(3)),
    MyGradientTracer(Set(4)),
]
```

```
ytracer = ▸ MyGradientTracer(Set([3, 2, 1]))
```

```
ytracer = f(xtracer)
```

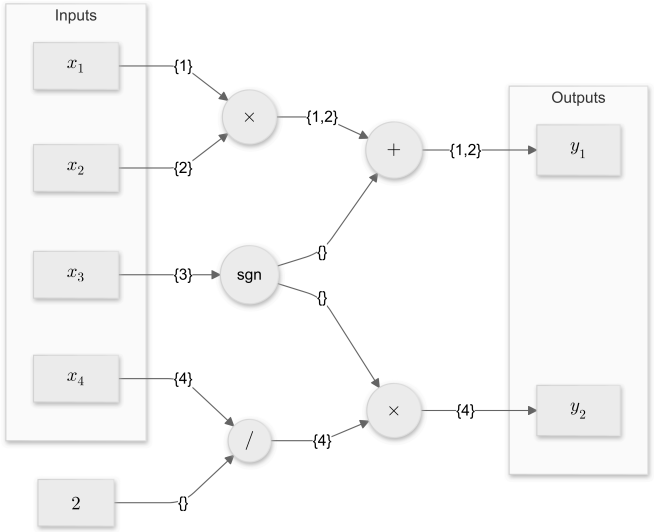
Sparsity detection with multiple outputs

```
F (generic function with 1 method)
```

```
1 F(x) = [x[1]*x[2]+sign(x[3]), sign(x[3]) * x[4]/2]
```

```
▸ [MyGradientTracer(Set([2, 1])), MyGradientTracer(Set([4]))]
```

```
1 F(xtracer)
```

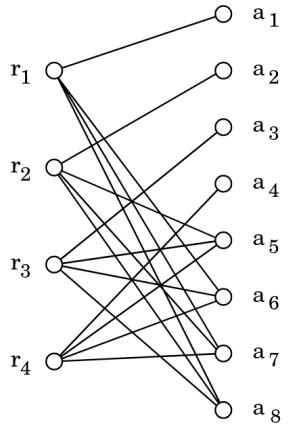


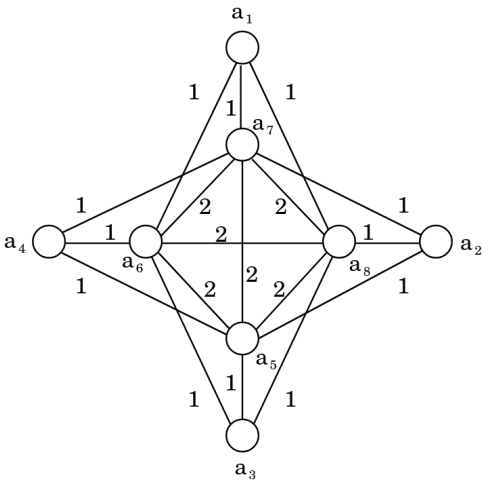
What color is your Jacobian ?

Given a sparse matrix A, two graphs represent its sparsity:

- Column intersection graph, used in Software for estimating sparse Jacobian matrices, 1984
- Bipartite graph, used in Colpack and SparseMatrixColoring

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 & a_{16} & a_{17} & a_{18} \\ 0 & a_{22} & 0 & 0 & a_{25} & 0 & a_{27} & a_{28} \\ 0 & 0 & a_{33} & 0 & a_{35} & a_{36} & 0 & a_{38} \\ 0 & 0 & 0 & a_{44} & a_{45} & a_{46} & a_{47} & 0 \end{bmatrix}$$





See Section 3.4 of What color is your Jacobian ?

Coloring problems

Given a graph $G(V, E)$,

- Nodes u, v are distance k neighbors if there exists a path from u to v of length at most k.
- A *distance-k* coloring : mapping $\phi : V \to \{1, \ldots, p\}$ such that $\phi(u) \neq \phi(v)$ whenever u, v are distance-k neighbors.
- *k-chromatic number* $\xi_k(G)$: minimum p such that $\exists$ distance-k coloring with p colors.
- Distance-k coloring problem : Find distance-k coloring with fewest colors.
- For every fixed integer $k \geq 1$, the distance-k graph coloring problem is NP-hard.

See Section 2.1, 2.2, 3.2 of What color is your Jacobian ?

Formulation as coloring problem

Theorem 3.5 A coloring of the columns is distance-2 in the bipartite graph iff columns of the same color are structurally orthogonal.

Let the k-th power G^k of a graph G be the graph with same nodes as G and an edge in G^k for each distance k neighbors in G.

Lemma 3.7 The column intersection graph is the square of the biparted graph restricted to its column vertices.

Lemma 2.1 A coloring is distance-k in G iff it is distance-1 in G^k.

Corollary A coloring of the columns is distance-1 in the column intersection graph iff columns of the same color are structurally orthogonal.

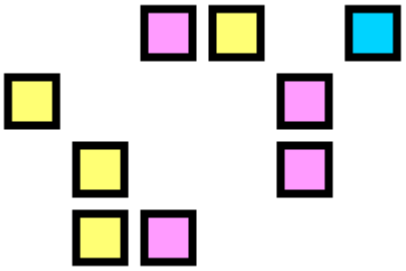
Example

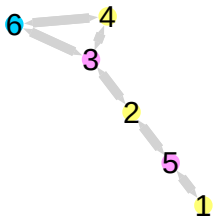
Consider the following sparsity pattern of the Jacobian matrix:

Taken from a tutorial of the SparseMatrixColorings' doc

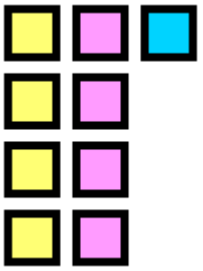
```
S = 4×6 SparseMatrixCSC{Int64, Int64} with 9 stored entries:
    ⋅  ⋅  1  1  ⋅  1
    1  ⋅  ⋅  ⋅  1  ⋅
    ⋅  1  ⋅  ⋅  1  ⋅
    ⋅  1  1  ⋅  ⋅  ⋅
```

Color's columns are structurally orthogonal





Jacobian from 3 JVP



What color is your Hessian ?

How many HVP do we need to find the values of this sparse **symmetric** matrix ?

```
ijkl = 4×4 SparseMatrixCSC{Int64, Int64} with 10 stored entries:
       1  2  3  ⋅
       2  4  ⋅  5
       3  ⋅  6  ⋅
       ⋅  5  ⋅  7
```

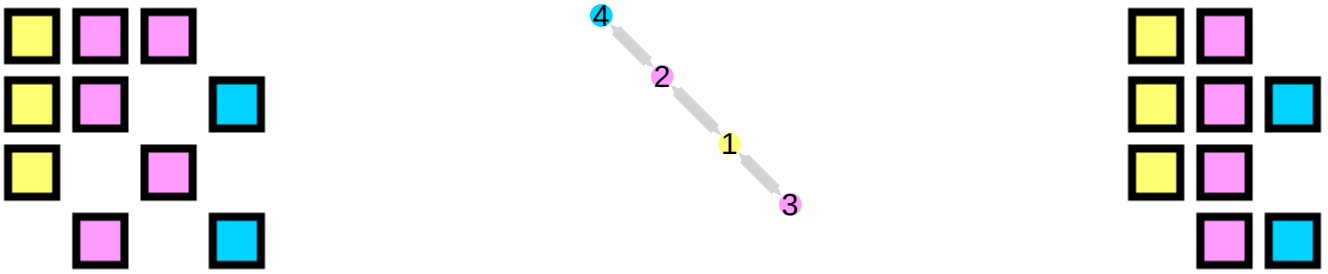
Section 2.4 Consider the *adjacency graph* G of a matrix A be the graph whose adjacency matrix has same sparsity pattern as A. So i and j are adjacent iff a_{ij} is nonzero.

▶ **Is a coloring of the columns is distance-2 in the adjacency graph iff columns of the same color are structurally orthogonal ?**

Do we need to aim for distance-2 coloring or does the symmetry of the matrix allows us to relax this ?

What changes with symmetry ?

$$D^\top = \begin{bmatrix} 1 & & & \\ & 1 & 1 & \\ & & & 1 \end{bmatrix} \qquad AD = \begin{bmatrix} a_1 & a_2 + a_3 & \\ a_2 & a_4 & a_5 \\ a_3 & a_6 & \\ & a_5 & a_7 \end{bmatrix}$$

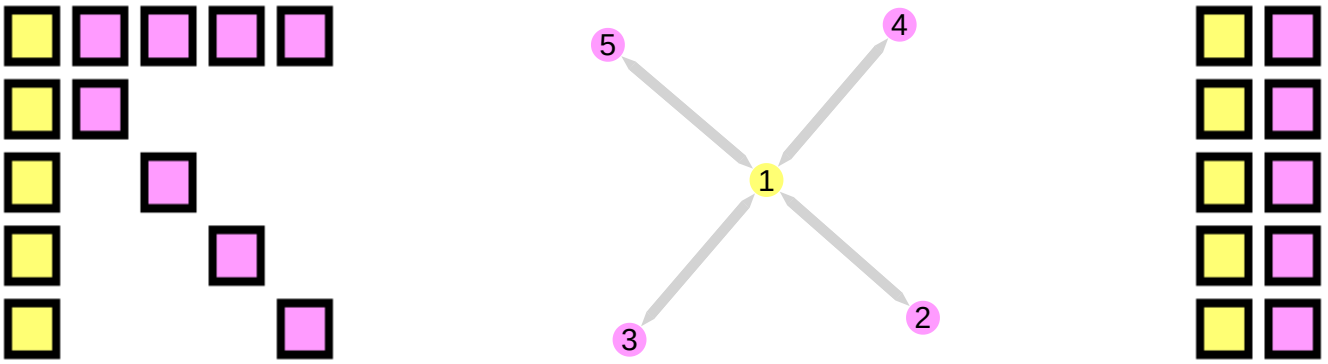


▶ **Is this a distance-2 coloring ? Can we deduce all values of the hessian given the values of AD.**

Definition 4.2 A coloring of the columns of a symmetric matrix A is *symmetrically orthogonal* if for every nonzero element a_{uw}, either (1) there is no other column of color $\phi(u)$ with a nonzero in row

r_u, or (2) there is no other column of color $\phi(v)$ with a nonzero in row r_v.

Coloring of a star



- For any pink node $u > 1$, the offdiagonal entry $a_{u,1}$ can be obtained from the yellow HVP.

n = 5

Star coloring

Definition 4.5 A mapping $\phi : V \to \{1, \ldots, p\}$ is a *star coloring* if

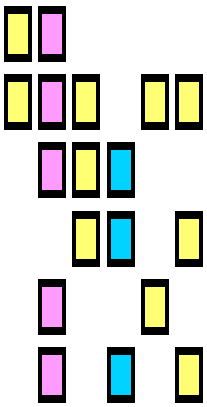
1. ϕ is a distance-1 coloring
2. every **path of 4 vertices** uses at least 3 colors

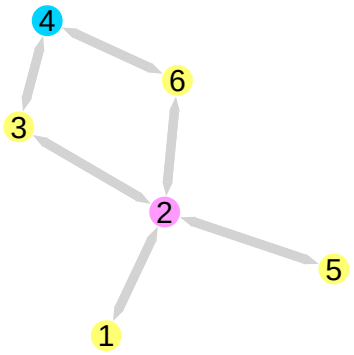
Theorem 4.6 Let A be a symmetric matrix with **nonzero diagonal elements**, G be its adjacency matrix. A mapping ϕ is a star coloring of the adjacency graph iff it induces a symmetrically structurally orthogonal partition of the columns of A.

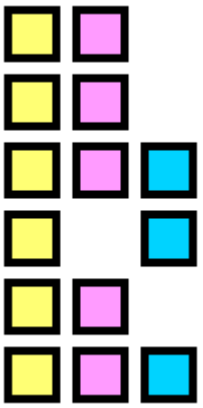
Name star coloring comes from the fact that the subgraph induced by any pair of colors is a star.

▶ **How is the subgraph induced by a pair of colors ?**

Small Example







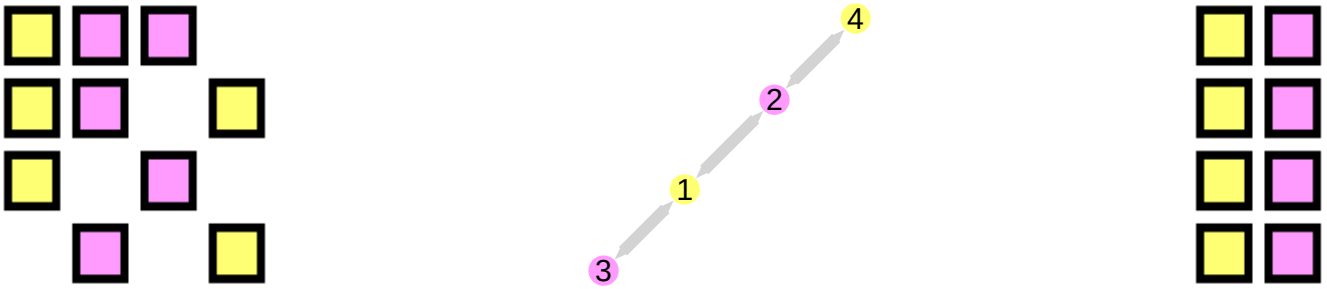
▶ **From which HVP do we determine the off-diagonal entries ?**

Less colors but nontrivial substitution

With star coloring, we need at least 3 colors. With 2 colors, our two forward tangents are

$$D^\top = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \qquad AD = \begin{bmatrix} a_1 & a_2 \\ a_2 + a_4 & a_3 \\ a_5 & a_4 + a_6 \\ a_6 & a_7 \end{bmatrix}$$

▶ **How do we get the value of a_4 ?**



Acyclic coloring

Definition 6.3 A mapping $\phi : V \to \{1, \dots, p\}$ is an *acyclic coloring* if

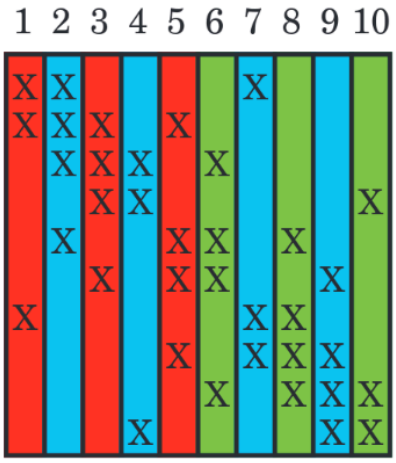
1. ϕ is a distance-1 coloring
2. every **cycle** uses at least 3 colors

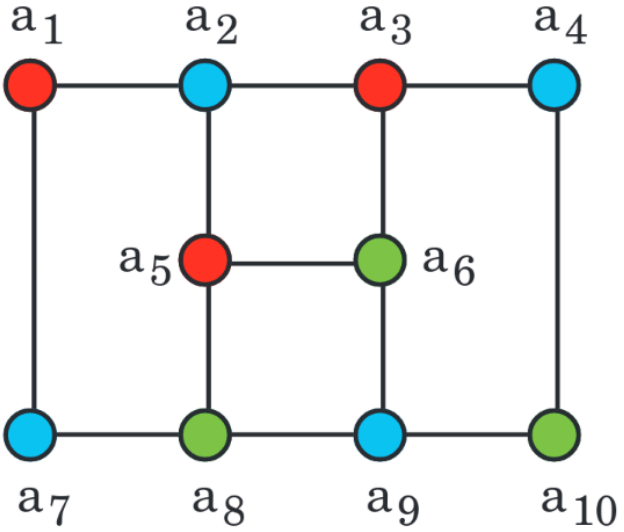
Theorem 4.6 Let A be a symmetric matrix with **nonzero diagonal elements**, G be its adjacency matrix. A mapping ϕ is an acyclic coloring of the adjacency graph iff it induces a substitutable partition of the columns of A.

Name acyclic coloring comes from the fact that the subgraph induced by any pair of colors is a forest.

▶ **How is the subgraph induced by a pair of colors ?**

Illustrative example





For each color c, we consider the submatrix $B_c = A_{:,I}$ where $I = \{i \mid \phi(i) = c\}$ is the set of columns corresponding to color-c nodes.

▶ **What are the nonzero entries of ith row of $B_{\phi(i)}$?**

Let h_c be the sum of the columns of B_c hence the result of the corresponding HVP.

▶ **How can the diagonal entries be determined ?**

Red-Blue subgraph

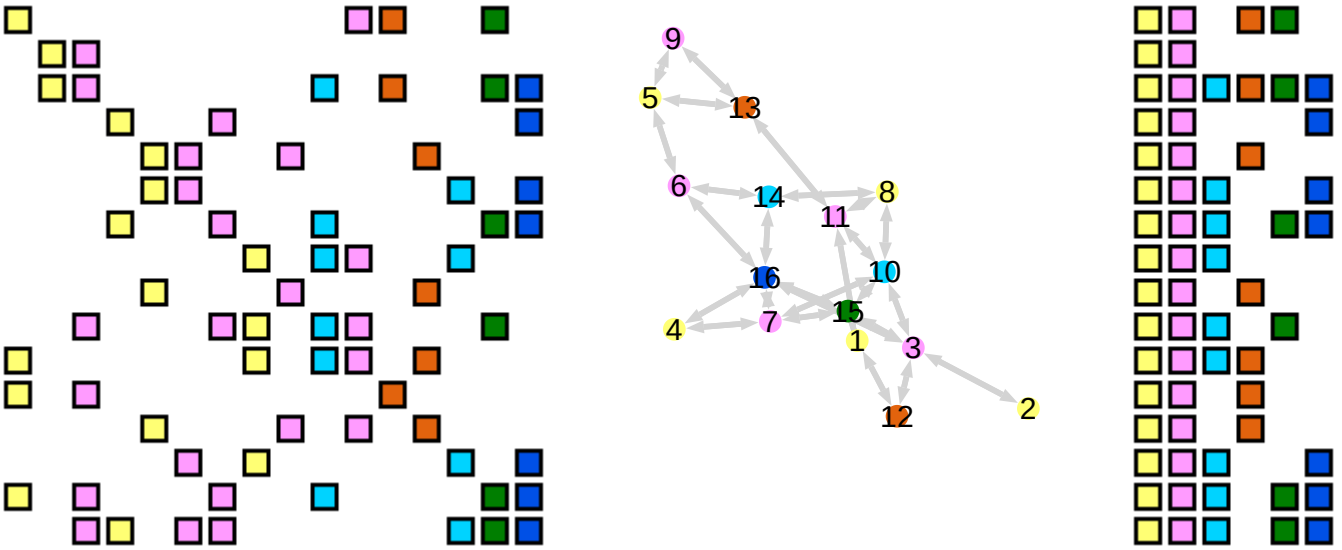
To compute an off-diagonal entry A_{ij}, consider the subgraph induced by the two colors $\phi(i)$ and $\phi(j)$. The entry will be computed, from $B_{\phi(i)}$ and $B_{\phi(j)}$, possible after substitution.

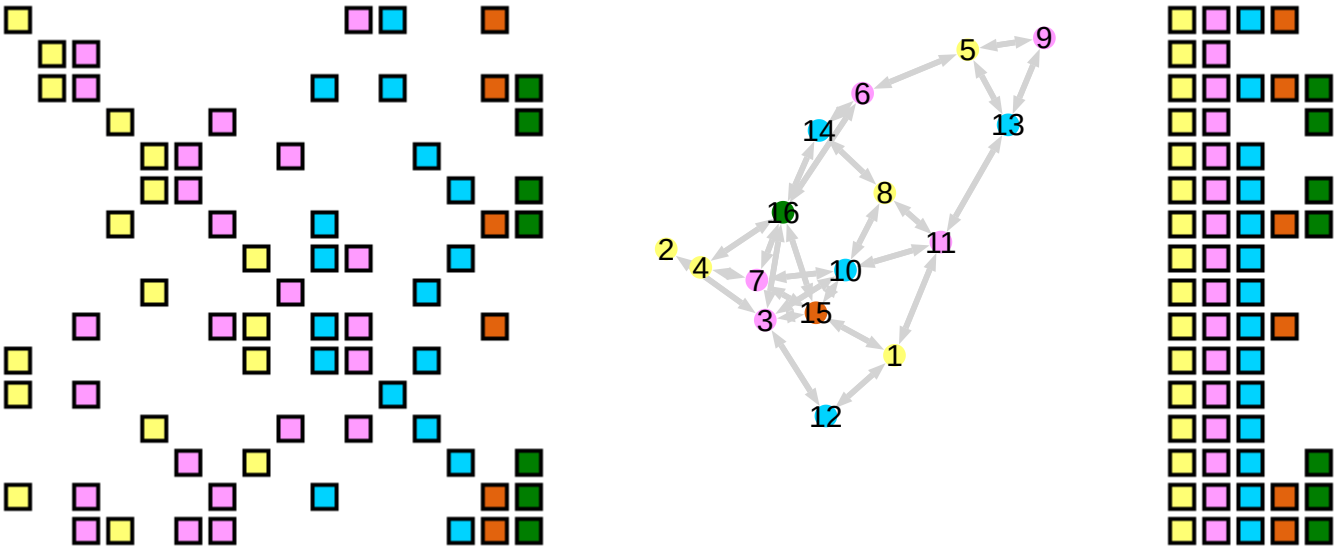
Consider below left B_{red} and right B_{blue}. The rows corresponding to diagonal entries have been omitted as we already found how to compute them in the previous slide. The rows corresponding to green nodes are in bold as they will be computed from the subgraph induced by red/green or blue/green so we can ignore them for now.

$$\begin{pmatrix} a_{21} & a_{23} & a_{25} \\ & a_{43} & \\ & \mathbf{a_{63}} & \mathbf{a_{65}} \\ a_{71} & & \\ & & \mathbf{a_{85}} \end{pmatrix}; \quad \begin{pmatrix} a_{12} & & a_{17} & \\ a_{32} & a_{34} & & \\ a_{52} & & & \\ & & & \mathbf{a_{69}} \\ & & \mathbf{a_{87}} & \mathbf{a_{89}} \\ & \mathbf{a_{10,4}} & & \mathbf{a_{10,9}} \end{pmatrix}$$

▶ **How to compute the non-bold entries ?**

Larger example : star vs acyclic coloring





n = 16

Comparison of chromatic numbers

Theorem 7.1 For every graph G,

$$\xi_1(G) \leq \xi_{\text{acyclic}}(G) \leq \xi_{\text{star}}(G) \leq \xi_2(G) = \xi_1(G^2)$$

▶ **What is the 1-chromatic number of planar graphs ?**

Utils

scale = pad = border =

```
1 using PlutoUI, PlutoUI.ExperimentalLayout, HypertextLiteral, PlutoTeachingTools
```

```
1 using SparseArrays, Images, SparseMatrixColorings
```

```
1 using Graphs, GraphPlot, LinearAlgebra, StableRNGs
```

```
viz (generic function with 1 method)
```

```
colored_plots (generic function with 1 method)
```

```
img (generic function with 3 methods)

two_columns (generic function with 1 method)

three_columns (generic function with 1 method)

qa (generic function with 2 methods)
```